

Soft Spaces in Low-Qubit Hilbert Spaces – Phase 3

Basis-Invariant Projector Geometry, Spectral $\pm E$ Organization,
Independent Replication, and Cross-Dimensional Confirmation

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Abstract

Phase 3 of the Soft Spaces project reformulates the high-qubit problem after the Phase-2 falsification audit showed that the earlier selected two-eigenvector hotspot score was not invariant under unitary rotations inside exactly degenerate eigenspaces. The central methodological change is to replace arbitrary eigenvector pairs by complete spectral subspaces and to define the perturbative response through projectors. For an exact eigenspace projector P , complement $Q = \mathbb{I} - P$, Hamiltonian H , perturbation V , reference energy E , and regularization $\eta > 0$, the working effective operator is

$$K_P^{(\eta)}(E) = PVQ \frac{E - QHQ}{(E - QHQ)^2 + \eta^2} QVP.$$

The corresponding normalized cancellation response is constructed from complete exact eigenspaces of QHQ , making the score invariant to arbitrary basis rotations inside degenerate eigenspaces. A frozen regression test confirmed invariance to numerical precision. Increasing the random-Pauli Hamiltonian from five to eleven terms reduced the extreme degeneracy of the earlier model and restored non-trivial subspace prominences.

The resulting spectrum displayed a strong organization between eigenspaces at energies $+E$ and $-E$. A sequence of frozen tests separated this dominant structural symmetry from the smaller REAL-specific component. At 8Q, an original 12-seed development set gave a positive difference

$$\Delta r = r_{\text{REAL}} - r_{\text{NULL}}$$

in 11 of 12 seeds, with a one-sided exact sign-test $p = 0.00317$, one-sided Wilcoxon $p = 0.00146$, and bootstrap intervals for both the mean and median entirely above zero. A fully independent, predeclared 12-seed replication then passed all five acceptance criteria: 11/12 positive seeds, mean $\Delta r = 0.0481$, median $\Delta r = 0.0404$, sign-test $p = 0.00317$, Wilcoxon $p = 0.000976$, and a 95% bootstrap confidence interval for the median of $[0.0242, 0.0568]$.

A subsequent predeclared dimensional replication tested the same endpoint independently at 7Q and 9Q. Both dimensions passed all five frozen criteria. At 7Q all 12 seeds were positive, with median $\Delta r = 0.1839$ and a 95% bootstrap interval $[0.0801, 0.2441]$. At 9Q all 12 seeds were again positive, with median $\Delta r = 0.01047$ and interval $[0.00326, 0.01947]$. The dominant $\pm E$ symmetry is therefore largely structural and shared with NULL, while a smaller REAL-specific enhancement is independently replicated and confirmed across Hilbert-space dimensions 7Q–9Q.

The Phase-3 result changes the interpretation of Soft Spaces. The evidence no longer supports isolated ordinal eigenvector hotspots as the primary physical object. Instead, the reproducible quantity is a basis-invariant organization of complete spectral subspaces, with a strong background symmetry and a smaller REAL-specific excess. A possible connection to projection-operator and open-quantum-system formalisms is identified, but no environmental or dissipative interpretation is claimed at this stage.

Keywords: Hilbert space; projector methods; Feshbach projection; spectral symmetry; random Pauli Hamiltonians; basis invariance; degenerate eigenspaces; REAL–NULL controls; perturbation theory; replication; Soft Spaces.

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1 Scientific aim of Phase 3

Phase 3 begins from a negative but decisive result of Phase 2. The earlier high-qubit continuation programme had produced recurrent candidate coordinates and a perturbative effective-subspace score. However, the final Phase-2 audit showed that when a selected two-dimensional slice lies inside a larger exactly degenerate eigenspace, the reported hotspot prominence can change under a legal unitary rotation of the eigenbasis while the Hamiltonian itself is unchanged. Such a statistic cannot be interpreted as a physical observable of H and V alone.

The Phase-3 problem is therefore not simply to search for more hotspots. It is to identify a physical object that survives representation changes. The guiding requirements are:

1. the primary object must be a complete subspace rather than an arbitrary basis slice;
2. the score must be invariant under unitary basis changes internal to that subspace;
3. the Hamiltonian ensemble must not be so sparse that massive spectator degeneracy dominates the geometry;
4. REAL and NULL must remain spectrum matched and be perturbed by the same physical operator families;
5. discovery, falsification, independent replication, and dimensional replication must be kept distinct;
6. no post-hoc seed or threshold tuning may be used to promote a result.

This leads naturally to a projector formulation.

2 From eigenvector pairs to complete spectral projectors

Let H be a finite-dimensional Hermitian Hamiltonian with spectral resolution

$$H = \sum_g E_g P_g, \quad P_g P_h = \delta_{gh} P_g, \quad \sum_g P_g = \mathbb{I}.$$

Here P_g is the projector onto the complete exact eigenspace associated with energy E_g . If the eigenspace has dimension $d_g > 1$, any orthonormal basis $\{|g, a\rangle\}_{a=1}^{d_g}$ may be used numerically, but the physical object is P_g , not the individual basis vectors.

For a target eigenspace $P = P_j$, define

$$Q = \mathbb{I} - P.$$

The Hilbert space decomposes as

$$\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}.$$

This is a projection-operator decomposition in the standard Feshbach sense [1, 2]. Importantly, at this stage Q means only the complementary Hilbert-space sector. It is not yet assumed to be an external environment.

3 Regularized second-order effective operator

For a Hermitian perturbation V , the unregularized second-order effective term has the familiar form

$$PVQ(E - QHQ)^{-1}QVP,$$

whenever the inverse exists. Since the numerical spectra contain near-resonant denominators, Phase 3 uses the odd regularized resolvent kernel

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}, \quad \eta > 0.$$

Thus the working operator is

$$\boxed{K_P^{(\eta)}(E) = PVQ g_\eta(E - QHQ) QVP.} \quad (1)$$

Using the exact spectral projectors P_g of H outside the target subspace,

$$Q = \sum_{g \neq j} P_g,$$

we obtain

$$K_P^{(\eta)}(E_j) = \sum_{g \neq j} r_{jg} PV P_g VP, \quad (2)$$

with

$$r_{jg} = \frac{E_j - E_g}{(E_j - E_g)^2 + \eta^2}. \quad (3)$$

Equation (2) is the central algebraic simplification of the Phase-3 score. Every contribution is expressed through projectors onto complete exact eigenspaces. If the basis within P is changed by a unitary U_P and the basis within P_g by U_g , then the matrix representation changes by unitary similarity, while all Frobenius norms appearing below remain invariant.

4 Basis-invariant cancellation score

Define

$$A_g = PVP_gVP.$$

The normalized cancellation score is

$$\boxed{C(P, V) = \frac{\left\| \sum_{g \neq j} r_{jg} A_g \right\|_F}{\sum_{g \neq j} |r_{jg}| \|A_g\|_F}.} \quad (4)$$

The denominator normalizes the coherent second-order sum against the magnitudes of the individual virtual channels. Values close to zero indicate strong cancellation between channels, while larger values indicate less cancellation. The precise physical interpretation remains under investigation; the essential Phase-3 property is that $C(P, V)$ is defined on the subspace projector and not on an arbitrary selected basis pair.

To see the invariance directly, let

$$P \mapsto U_P P U_P^\dagger = P, \quad P_g \mapsto U_g P_g U_g^\dagger = P_g$$

for unitaries acting only inside the respective exact eigenspaces. Then the matrix of A_g in the target subspace changes as

$$A_g \mapsto U_P^\dagger A_g U_P,$$

and therefore

$$\|A_g\|_F = \|U_P^\dagger A_g U_P\|_F.$$

The same applies to the sum in the numerator, proving basis invariance of the score under internal eigenspace rotations.

5 REAL–NULL construction and robust local prominence

The REAL model uses the eigenbasis of the generated Hamiltonian. The NULL model is spectrum matched: it preserves the same eigenvalues but replaces the eigenbasis by a deterministic Haar-random unitary. Thus

$$H_{\text{NULL}} = U_{\text{Haar}} \text{diag}(E_1, \dots, E_D) U_{\text{Haar}}^\dagger.$$

The same perturbation operator V_f is applied to REAL and NULL for each frozen family f . In Phase 3 the two perturbation families are dephasing-like Z/ZZ and transverse X/XX operators.

For each eigenspace P_j define the family-specific contrast

$$\Delta C_f(j) = C_{\text{REAL}}(P_j, V_f) - C_{\text{NULL}}(P_j, V_f).$$

The conservative robust contrast is

$$R(j) = \min_f \Delta C_f(j). \quad (5)$$

To remove broad spectral background, define a local neighborhood $\mathcal{N}(j)$ in energy-order index and the local prominence

$$\Pi(j) = R(j) - \text{median}_{k \in \mathcal{N}(j)} R(k). \quad (6)$$

The Phase-3 discovery problem is therefore no longer “which eigenvector index is special?” but rather “which complete spectral subspaces display a locally unusual REAL–NULL projector response?”

6 The Phase-3 working equation

The mathematical chain developed in Phase 3 can be summarized as

$$\boxed{(H, V) \longrightarrow P \longrightarrow K_P^{(\eta)} \longrightarrow C(P, V) \longrightarrow \Delta C(P) \longrightarrow \mathcal{S}_{\pm E}.} \quad (7)$$

The final quantity is defined from the correlation of the prominence field across spectrally opposite eigenspaces. Let P_{+E} and P_{-E} denote exact eigenspaces with opposite energies. Define

$$r_{\text{REAL}} = \text{Corr}_{E, -E} [\Pi_{\text{REAL}}(E), \Pi_{\text{REAL}}(-E)]$$

and analogously r_{NULL} . The Phase-3 working order parameter is

$$\boxed{\mathcal{S}_{\text{SS}} \equiv \Delta r = r_{\text{REAL}} - r_{\text{NULL}}.} \quad (8)$$

Equation (8) should be regarded as a working empirical order parameter, not as a fundamental law. Its value is that it survived independent replication and cross-dimensional confirmation after the earlier basis-dependent formulation had been rejected.

7 Falsification-to-reformulation sequence

The Phase-3 development proceeded through a deliberately adversarial sequence.

7.1 v30.0: projector-score invariance

The first test replaced the arbitrary pair score by the degeneracy-closed projector score while preserving the earlier 8Q five-term environment. Legal Haar rotations were applied independently inside every exact degenerate eigenspace. The reconstructed Hamiltonians remained unchanged to approximately 10^{-14} – 10^{-15} , while the new prominence was invariant at approximately 10^{-16} . The basis-dependence counterexample was therefore removed numerically.

The same run also exposed a second problem: the five-term Hamiltonian possessed only four exact energy groups of dimension 64 in the smoke seed. Once complete eigenspaces were treated correctly, the local prominence collapsed to numerical zero. The sparse Hamiltonian was therefore too degenerate to support the intended local spectral interpretation.

7.2 v30.1: Hamiltonian-complexity scan

The number of random Pauli terms was scanned through 5, 7, 9, and 11 while the remaining logic was frozen. In the first seed the exact group structure changed as

$$4 \times 64 \longrightarrow 16 \times 16 \longrightarrow 16 \times 16 \longrightarrow 64 \times 4.$$

The eleven-term ensemble was adopted as the primary Phase-3 working model because it substantially reduced the artificial degeneracy without changing the projector formalism.

7.3 v30.2: complete-eigenspace discovery scan

The obsolete adjacent-pair eligibility gate was removed and complete exact eigenspaces were scanned directly. Across the original twelve 8Q seeds, the eleven-term model produced non-trivial local prominences in every seed. The strongest smoke-seed candidates appeared at approximately opposite energies, for example

$$E = +0.635388301 \quad \text{and} \quad E = -0.635388301,$$

with prominences approximately 0.633 and 0.598. Similar $\pm E$ pairing occurred repeatedly throughout the seed ensemble.

8 Separating hotspot amplitude from spectral organization

A critical distinction emerged in v30.4. The absolute maximum hotspot amplitude was compared with shuffle-null distributions, while the $\pm E$ organization was tested independently. For the full twelve-seed 8Q set, no seed had a significant maximum-hotspot amplitude at the 5% level, whereas all twelve seeds had significant $\pm E$ correlation and significant paired-strong enrichment.

Thus the evidence did *not* support the statement that REAL merely produces unusually large isolated maxima. Instead, the reproducible structure was relational: responses at $+E$ and $-E$ were highly organized.

The decomposition test v30.5 then showed that this organization was not uniquely REAL. The median correlations across the first twelve 8Q seeds were

$$\text{median}(r_{\text{REAL}}) = 1.000000, \quad \text{median}(r_{\text{NULL}}) = 0.967896, \quad \text{median}(r_{\text{REAL}} - r_{\text{NULL}}) = 0.032104.$$

The dominant symmetry is therefore shared with NULL and is most naturally attributed to the Hamiltonian/spectral structure. The scientifically relevant residual is the smaller excess Δr .

9 Across-seed evidence at 8Q

The v30.6 aggregate analysis used the original twelve frozen 8Q seeds. The difference Δr was positive in 11 of 12 seeds. The primary robust statistics were

$$\begin{aligned}\text{mean}(\Delta r) &= 0.050466, \\ \text{median}(\Delta r) &= 0.032104, \\ p_{\text{sign,one-sided}} &= 0.00317383, \\ p_{\text{Wilcoxon,one-sided}} &= 0.00146449.\end{aligned}$$

The 95% bootstrap confidence intervals were

$$\text{mean} : [0.022297, 0.084095], \quad \text{median} : [0.012663, 0.063867].$$

The raw paired standardized effect size was $d_z = 0.878$.

These results motivated a fully independent confirmation rather than further tuning.

10 Independent predeclared replication at 8Q

Version v30.7 used twelve new Hamiltonian seeds, 25043000 through 25043011, and froze five acceptance criteria before execution:

- C1. $\text{median}(\Delta r) > 0$;
- C2. at least 9 of 12 seeds have $\Delta r > 0$;
- C3. one-sided exact sign-test $p < 0.05$;
- C4. one-sided Wilcoxon signed-rank $p < 0.05$;
- C5. the 95% bootstrap interval for the median lies entirely above zero.

All five criteria passed. Table 1 summarizes the replication.

Table 1: Independent 8Q replication (v30.7).

Quantity	Result
Positive Δr seeds	11/12
Mean Δr	0.048107
Median Δr	0.040406
Exact sign test, one-sided	$p = 0.00317383$
Wilcoxon, one-sided	$p = 0.00097632$
95% bootstrap CI, mean	[0.025479, 0.074397]
95% bootstrap CI, median	[0.024192, 0.056815]
Cohen d_z	1.056863
Predeclared decision	CONFIRMED

The agreement between the development set and the independent replication is notable: the mean effect changed only from approximately 0.0505 to 0.0481, while the median changed from 0.0321 to 0.0404. This is substantially stronger evidence than re-analysis of the original seeds alone.

11 Cross-dimensional replication: v30.8

The next predeclared question was whether the 8Q result is dimension specific. Version v30.8 therefore tested 7Q and 9Q using twelve new independent seeds per dimension, 25044000 through 25044011, with the same five acceptance criteria applied separately to each dimension.

Table 2: Aggregate dimensional replication results.

Quantity	7Q	8Q replication	9Q
Positive seeds	12/12	11/12	12/12
Mean r_{REAL}	0.984588	0.998061	0.999795
Mean r_{NULL}	0.792616	0.949954	0.984218
Mean Δr	0.191973	0.048107	0.015577
Median Δr	0.183939	0.040406	0.010472
Sign test, one-sided	0.000244	0.003174	0.000244
Wilcoxon, one-sided	0.000488	0.000976	0.000488
95% CI median	[0.0801,0.2441]	[0.0242,0.0568]	[0.00326,0.01947]
Cohen d_z	1.343	1.057	0.797
Decision	CONFIRMED	CONFIRMED	CONFIRMED

Both 7Q and 9Q passed all five frozen criteria. Therefore the predeclared dimensional decision was

CONFIRMED ACROSS DIMENSIONS.

Figure 1 shows the observed decline in the REAL-specific excess from 7Q to 9Q.

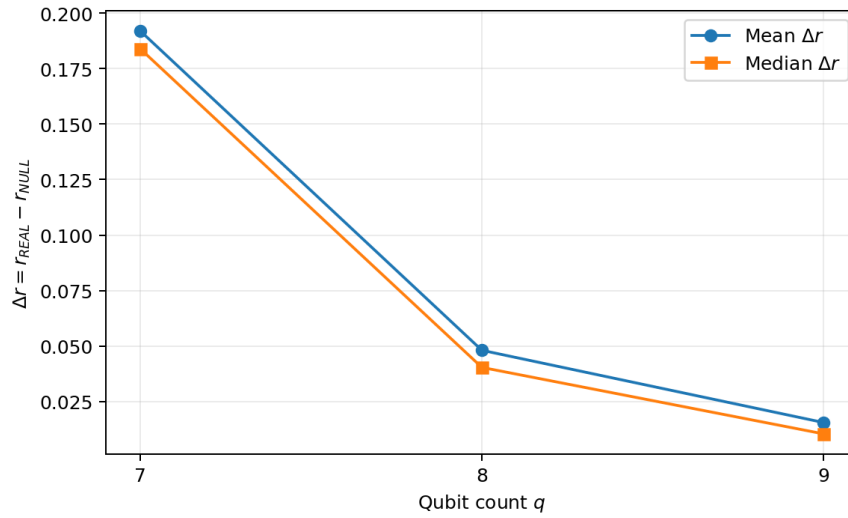


Figure 1: Observed mean and median $\Delta r = r_{\text{REAL}} - r_{\text{NULL}}$ at 7Q, the independent 8Q replication, and 9Q. The monotonic decline is an empirical observation, not yet a fitted scaling law.

One 7Q seed is especially informative. Seed 25044004 produced 128 eigenspaces of size 1, hence no exact degeneracy, yet still gave

$$r_{\text{REAL}} = 0.985464, \quad r_{\text{NULL}} = 0.473394, \quad \Delta r = 0.512070.$$

This demonstrates that the replicated REAL-specific effect does not require degenerate eigenspaces to exist.

12 Physical interpretation

The Phase-3 evidence supports a layered interpretation.

12.1 The old ordinal hotspot is not the physical object

The Phase-2 basis-rotation falsification showed that a selected two-vector slice inside a larger degenerate eigenspace is representation dependent. Phase 3 therefore rejects the ordinal pair itself as the primary physical object.

12.2 The complete spectral subspace is the natural object

The complete projector P_j is invariant under internal basis rotations. The second-order operator $K_P^{(\eta)}$ and the normalized response $C(P, V)$ are therefore attached to a subspace rather than to an arbitrary numerical eigenbasis.

12.3 The dominant $\pm E$ symmetry is structural

NULL exhibits very high $\pm E$ correlations. At 8Q the median NULL correlation was approximately 0.968, and in the independent replication the mean NULL correlation was approximately 0.950. Consequently, the near-perfect spectral pairing cannot be claimed as a uniquely REAL phenomenon.

12.4 A smaller REAL-specific excess remains

After subtraction at the level of the $\pm E$ organization, the residual Δr remains positive with strong across-seed consistency. This residual survived an independent 8Q replication and separate 7Q and 9Q confirmation sets under frozen gates.

The strongest defensible statement is therefore:

The dominant $\pm E$ organization is largely a property shared by REAL and spectrum-matched NULL systems, but REAL exhibits a smaller, statistically reproducible enhancement that persists under independent seed replication and across 7Q–9Q Hilbert-space dimensions.

13 A provisional dimensional hypothesis

The dimensional results suggest a new, explicitly provisional hypothesis. Across 7Q, 8Q, and 9Q, r_{REAL} remains near unity while r_{NULL} rises toward unity. Consequently the excess

$$\Delta r(q) = r_{\text{REAL}}(q) - r_{\text{NULL}}(q)$$

decreases rapidly with q .

The observed medians are approximately

$$\Delta r(7) = 0.1839, \quad \Delta r(8) = 0.0404, \quad \Delta r(9) = 0.0105.$$

With only three dimensions it would be inappropriate to fit or promote a scaling law. Nevertheless, the data motivate the qualitative hypothesis

$$\boxed{r_{\text{REAL}}(q) \approx 1, \quad r_{\text{NULL}}(q) \rightarrow 1, \quad \Delta r(q) \downarrow 0 \quad \text{as } q \text{ increases.}} \quad (9)$$

A future dimensional programme can test whether this decrease is exponential, algebraic, ensemble specific, or eventually saturating.

14 Relation to projection methods and open quantum systems

Equation (1) is structurally close to effective-Hamiltonian and projection-operator constructions used in many-body and open-system theory. Feshbach projection divides the state space into P and Q sectors and eliminates Q to obtain an effective operator acting in P [1, 2]. Related projection methods underlie the Nakajima–Zwanzig formalism for reduced dynamics [3, 4, 5].

This resemblance is physically suggestive but must not be overstated. In the current Phase-3 model,

$$Q = \mathbb{I} - P$$

is simply the complement of the selected spectral subspace. It is not a separately modeled bath, reservoir, or environment. No partial trace over Q , no dissipator, and no master equation is used in the Phase-3 evidence presented here.

A genuine open-system extension would require an explicit system–environment partition, for example

$$H_{\text{tot}} = H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E + \lambda H_{SE},$$

followed by a reduced state

$$\rho_S(t) = \text{Tr}_E \rho_{SE}(t).$$

Only then could one ask whether the projector structures identified here correspond to unusually small decay rates, suppressed decoherence, protected sectors, or cancellation in self-energy channels. The present result should therefore be viewed as a mathematically natural bridge to open-system questions, not as an open-system demonstration.

15 Limitations

Several limitations remain important.

1. **Model dependence.** The current evidence uses an eleven-term random-Pauli Hamiltonian ensemble. Generalization to other local, sparse, dense, or physically motivated Hamiltonians has not yet been established.
2. **Symmetry mechanism unresolved.** The near-perfect $\text{REAL} \pm E$ correlation suggests an algebraic symmetry, potentially of chiral or anticommuting type, but no operator S satisfying $SHS^{-1} = -H$ has yet been derived for the ensemble.
3. **No fundamental scaling law.** The decline of Δr from 7Q to 9Q is real in the present data but is based on only three dimensions including the prior 8Q replication.
4. **NULL saturation.** The approach of r_{NULL} toward unity may reflect high-dimensional concentration or a structural consequence of the Pauli ensemble. This requires direct analysis.
5. **Open-system interpretation is hypothetical.** The P/Q notation alone does not make Q an environment.
6. **Finite numerical ensembles.** The replication design is strong for the stated seed blocks but does not constitute a theorem over the full random ensemble.

16 Phase 3.2 programme

The evidence base is now sufficiently stable that the next stage should focus on mechanism rather than further parameter tuning. A natural Phase-3.2 sequence is:

1. identify the algebraic origin of the $\pm E$ symmetry and test for an exact or approximate operator relation such as

$$SHS^{-1} = -H;$$
2. determine how this symmetry acts on P_g , V , and the effective operator $K_P^{(\eta)}$;
3. analytically separate the symmetry-forced component of r_{REAL} and r_{NULL} from the residual REAL-specific term;
4. test the dimensional behavior of $\Delta r(q)$ on additional qubit counts only after the mechanism-level predictions are frozen;
5. subsequently investigate whether an explicit open-system model maps the projector response to self-energy, leakage, decoherence, or lifetime observables.

This sequence preserves the falsification-driven logic of the project: first identify what is structural, then isolate what remains, then seek a physical mechanism.

17 Conclusion

Phase 3 converts the central Soft Spaces question from a basis-index search into a basis-invariant subspace problem. The earlier two-eigenvector formulation failed a decisive degeneracy-rotation test. Replacing it by a complete-projector score removed that ambiguity. A richer eleven-term Hamiltonian reduced the massive degeneracy of the five-term model and restored non-trivial local subspace structure.

The resulting $\pm E$ organization is extremely strong but is largely shared with spectrum-matched NULL controls. The primary Phase-3 result is therefore not the symmetry itself but the residual

$$\Delta r = r_{\text{REAL}} - r_{\text{NULL}}.$$

This residual was positive in 11 of 12 development seeds, independently replicated in 11 of 12 new 8Q seeds under five predeclared criteria, and then confirmed in all 12 independent seeds at both 7Q and 9Q under the same gate.

The evidence therefore supports a modest but substantive claim: a small REAL-specific enhancement of basis-invariant $\pm E$ subspace organization survives rigorous control, independent replication, and dimensional replication from 7Q through 9Q. At the same time, the data show that the dominant symmetry is structural rather than uniquely REAL. This distinction is central. Phase 3 does not restore the old hotspot claim; it replaces it with a narrower and better-defined physical question.

The remaining mystery is now substantially sharper: why does the REAL projector response approach perfect $\pm E$ organization, and why does the NULL background approach the same organization as Hilbert-space dimension grows? Answering that question is the natural starting point for Phase 3.2.

A Statistical decision rules

For $n = 12$ independent seedwise differences Δr_s , the exact sign test uses

$$K = \sum_{s=1}^{12} \mathbf{1}(\Delta r_s > 0),$$

with the null model

$$K \sim \text{Binomial}(12, 1/2).$$

The one-sided tail probability is

$$p_{\text{sign}} = \sum_{k=K}^{12} \binom{12}{k} 2^{-12}.$$

For $K = 11$ this gives

$$p_{\text{sign}} = \frac{13}{4096} \approx 0.00317383,$$

and for $K = 12$,

$$p_{\text{sign}} = \frac{1}{4096} \approx 0.00024414.$$

The Wilcoxon signed-rank test retains both sign and rank magnitude information and was used as a complementary non-parametric aggregate test. Bootstrap confidence intervals were generated by resampling the twelve seedwise Δr values with replacement and recomputing the mean or median.

B Predeclared dimensional gate

For each independently tested dimension q , v30.8 required all five conditions:

- C1 : $\text{median}_s \Delta r_s > 0$,
- C2 : $\#\{s : \Delta r_s > 0\} \geq 9$,
- C3 : $p_{\text{sign, one-sided}} < 0.05$,
- C4 : $p_{\text{Wilcoxon, one-sided}} < 0.05$,
- C5 : $\text{CI}_{0.95}^{\text{bootstrap}}(\text{median } \Delta r) \subset (0, \infty)$.

The overall dimensional claim required both 7Q and 9Q to pass independently.

References

- [1] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* **5**, 357–390 (1958).
- [2] H. Feshbach, “A unified theory of nuclear reactions. II,” *Annals of Physics* **19**, 287–313 (1962).
- [3] S. Nakajima, “On quantum theory of transport phenomena,” *Progress of Theoretical Physics* **20**, 948–959 (1958).
- [4] R. Zwanzig, “Ensemble method in the theory of irreversibility,” *Journal of Chemical Physics* **33**, 1338–1341 (1960).
- [5] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002).